Multitask Assessment of Multilingual Email Writing

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Abstract

This report presents a multilingual multitask system for predicting grammatical quality, content coverage, and CEFR proficiency from an email-writing prompt and a candidate response. The system combines controlled synthetic examples with authentic CEFR-labelled learner texts and fine-tunes XLM-R using parameter-efficient adaptation.

1 Introduction and Motivation

Assessing a written response requires more than detecting spelling or grammatical errors. A response may be grammatically accurate while failing to address the requested task, or it may contain all the required information while using limited vocabulary and simple sentence structures. Similarly, a short email is not necessarily written by a beginner, and a longer email is not necessarily evidence of advanced language proficiency. These distinctions make automated writing assessment a naturally multitask problem.

The objective of this work is to predict three properties of a candidate email:

1. a grammatical-quality score from 0 to 5;
2. a content-coverage score from 0 to 4; and
3. a CEFR proficiency level from A1 to C2.

The input consists of an email-writing prompt and the corresponding candidate response. The system must operate across Italian, Dutch, Greek, Polish, and Portuguese. This introduces two simultaneous challenges. First, the model must interpret both the instructions in the prompt and the linguistic quality of the response. Second, it must apply a consistent scoring interpretation across five languages with different morphology, agreement systems, word-order constraints, and orthographic conventions.

The three targets are related, but they are not interchangeable. Grammar measures the linguistic correctness and readability of the response. Content measures how completely the response addresses the information requested by the prompt. CEFR represents a broader notion of proficiency that includes vocabulary, syntactic range, cohesion, precision, fluency, register, and the ability to communicate nuanced ideas. Consequently, grammar and content scores provide useful auxiliary signals, but neither should determine CEFR on its own.

For example, a writer may produce a grammatically accurate email that mentions only one of several requested details. Such a response should receive a relatively high grammar score but a low content score. Conversely, a response may include every requested fact while containing agreement, case, conjugation, or word-order errors. It may therefore receive a high content score and a lower grammar score. A short but precise and contextually appropriate message can also demonstrate advanced proficiency. These cases motivated us to avoid a fixed rule in which grammar, content, email length, and CEFR always increase together.

A central difficulty was the absence of a single training corpus that matched the complete task. Public CEFR corpora contain useful examples of learner proficiency, but they do not generally provide the email-specific prompts, content-coverage scores, and grammar scores required here. Conversely, controlled synthetic emails make it possible to assign explicit grammar and content labels, but synthetic data can contain repeated structures and artificial correlations that do not fully reflect naturally occurring learner writing.

We therefore constructed a hybrid training resource. Its first part contains controlled multilingual email responses derived from a bank of writing scenarios. Each scenario was represented through explicit task components, allowing content coverage to be varied without changing the underlying prompt. Language-specific perturbation rules were then used to create grammatical errors with recorded types and severities. CEFR labels were assigned only to examples considered sufficiently unambiguous, while uncertain examples retained their grammar and content supervision but were excluded from the CEFR loss.

The second part of the training resource contains authentic CEFR-labelled learner texts obtained from public corpora. These texts do not provide email prompts or the two rubric scores, so they supervise only the CEFR objective. Where a source assigned the same response to an adjacent CEFR band, the uncertainty was retained as an interval rather than forcing an arbitrary exact label. This combination allowed the model to learn controlled task-specific scoring signals from the synthetic emails and broader proficiency patterns from naturally occurring learner writing.

For modelling, we used XLM-RoBERTa as a shared multilingual encoder and adapted it using low-rank adaptation. A common representation feeds two bounded regression heads for grammar and content and an ordinal head for CEFR. Missing labels are masked independently, which permits examples with different kinds of supervision to participate in the same training procedure.

The main contributions of this work are therefore:

- a controlled multilingual email dataset covering five languages, multiple communicative domains, and independently varied grammar and content scores;

- a typed, language-aware error-generation procedure that records each introduced grammatical or orthographic error;
- a hybrid CEFR supervision strategy combining conservative synthetic anchors, advanced email variants, exact corpus labels, and adjacent-level interval labels;
- a parameter-efficient multitask architecture that jointly predicts grammar, content, and ordinal CEFR proficiency; and
- an analysis of internal validation performance, pooling choices, calibration, and the generalization gap observed on the supplied evaluation data.

The remainder of this report describes the dataset construction, preprocessing pipeline, model architecture, training objectives, experimental results, and limitations of the approach.

2 Dataset Construction

2.1 Construction Strategy

The training data had to support three different objectives. Grammar required examples containing identifiable linguistic errors, content required responses with controlled levels of task completion, and CEFR required examples spanning general proficiency from A1 to C2. No single available corpus provided all three annotations for the five target languages.

We therefore used two complementary sources of supervision:

1. a controlled multilingual email dataset constructed for the task, containing grammar, content, and selected CEFR labels; and
2. authentic learner texts from public CEFR corpora, containing CEFR supervision but no task-specific grammar or content labels.

The controlled portion made it possible to define exactly which task details were present and which errors had been introduced. The authentic portion was added to reduce dependence on synthetic writing patterns and to expose the model to naturally occurring learner language. The final hybrid dataset contained 4,000 examples.

2.2 Clean Multilingual Scenario Bank

We first constructed 40 email-writing scenarios. The scenarios covered seven broad communicative domains, including workplace communication, customer service, education, healthcare, travel and accommodation, financial or administrative communication, and community-related communication. A more specific subcategory was also retained for each scenario. Examples included rescheduling a meeting, reporting a security incident, requesting a repair, correcting meeting minutes, renewing library material, and submitting insurance documents.

Each scenario was localized into all five target languages:

- Italian,
- Dutch,
- Greek,
- Polish, and
- Portuguese.

This produced

$$40 \text{ scenarios} \times 5 \text{ languages} = 200 \text{ clean multilingual bases.}$$

The localizations were stored explicitly rather than being translated dynamically during dataset generation. The initial wording was produced with language-model assistance and then manually inspected for consistency of facts, identifiers, dates, requested actions, and register. The term `manually_curated_multilingual` in the stored metadata refers to this inspection and explicit storage process. It should not be interpreted as professional translation by independent native-language annotators.

Each scenario-language pair contained an email prompt and a structured response template. The response was divided into the following fields:

- `GREETING`,
- `PURPOSE`,
- `CONTEXT`,
- `ACTION`,
- `SUPPORTING`,
- `CONFIRMATION`, and
- `CLOSING`.

The five middle fields carried the task-relevant information. The greeting and closing controlled the surface form of the email but were not counted as content details. This representation allowed individual parts of a response to be removed or retained without rewriting the entire email.

For example, a security-incident scenario could be separated into the following information:

- the purpose of reporting suspicious account access;
- the context that the writer did not perform the access and had changed the password;
- the requested investigation;
- the attached security screenshot; and
- a request to confirm that the incident had been registered.

The same factual structure was maintained across the five language versions.

2.3 Scenario Metadata and Dataset Split

Each scenario was assigned a macro category and a more specific subcategory. The purpose of this distinction was to retain the precise scenario description while also enabling broader checks of domain coverage.

The train-validation division was performed at the scenario level, not at the individual email level. Of the 40 scenarios, 32 were assigned to training and eight to validation. All five language versions and all derived variants of a scenario remained in the same split. Consequently, a model could not observe the Italian version of a scenario during training and then be evaluated on its Dutch or Greek version.

The split was additionally checked to ensure that every broad category represented in validation was also represented in training. This prevented the validation score from depending entirely on domains that were absent from the training set.

2.4 Atomic Task Details

A response component can contain more than one fact. For example, a single context sentence may mention both the date of an incident and an action already taken by the writer. We therefore created an additional detail schema containing between four and six atomic details for each scenario.

Every atomic detail had:

- a unique detail identifier;
- a link to its source component; and
- an importance weight of either 1 or 2.

These annotations were retained for traceability and auditing. They made it possible to inspect which facts were lost when a component was removed. However, the final content labels were generated using the number of broad task components present, rather than directly summing the atomic-detail weights. This distinction is important because the detail weights were metadata for analysis, not the final content-score formula.

2.5 Surface-Form Variation

Using a single prompt template, greeting, and closing for every example would create shortcuts unrelated to writing quality. We therefore introduced surface variation independently of the target scores.

Five prompt styles were created for each clean prompt:

1. the original direct instruction;
2. a first alternative imperative;
3. a second alternative imperative;
4. explicit task framing; and
5. situation-based framing.

Table 1: Controlled generation of content scores.

Score	Components	Construction
0	0	Generic or off-topic response
1	1	Purpose only
2	3	Purpose and two other components
3	4	Purpose and three other components
4	5	All task components

For example, the Italian equivalents of "write an email", "prepare an email", and "your task is to write an email" expressed the same requirements using different openings. Corresponding alternatives were defined separately for each language.

Pools of greetings and closings were also created for every language. Most scenarios used a formal register, while a small set of suitable scenarios, such as volunteering, housing, hotel, and library communication, could use a semi-formal register. The selected prompt style, greeting, and closing were sampled reproducibly and independently of the target score.

The recipient and writer were represented separately. The placeholder `[RECIPIENT]` was used in personalized greetings, while `[NAME]` remained the writer's signature. This avoided using the same placeholder for two different communicative roles.

2.6 Content-Score Generation

For each of the 200 clean multilingual bases, five content variants were generated. This produced 1,000 clean prompt-response pairs before grammatical perturbation.

The operational content-scoring rule is shown in Table 1.

The `PURPOSE` component was retained for scores 1 to 4 because a minimally relevant response should indicate why the writer is contacting the recipient. For scores 2 and 3, the remaining components were selected using a reproducible random procedure. Thus, a lower content score did not always mean that the attachment or confirmation sentence had been removed. Across the dataset, context, action, supporting information, and confirmation were all omitted at different rates.

Content score 0 represented responses that did not adequately perform the task. These emails were deliberately generic, off-topic, or too short to communicate the requested information. Examples included statements such as asking for help with "a matter" without identifying the actual issue.

Neutral opening and ending sentences were drawn from language-specific pools. These included expressions such as hoping that the recipient was well, thanking the recipient for their time, or remaining available if needed. They were not counted as task components. Similar filler sentences were distributed across several content levels so that one fixed phrase would not identify a particular score.

Responses with content scores from 1 to 4 were required to contain at least three body sentences. Content-zero responses remained shorter. This rule separated the rubric's very-short-response condition from ordinary in-

Table 2: Grammar-score distribution in the controlled dataset.

Grammar score	Number of examples
0	200
1	266
2	265
3	269
4	400
5	400

complete answers. It did not imply that a longer response automatically received a higher content score.

For every generated example, the dataset stored the selected content score, present and omitted components, present and omitted detail identifiers, prompt style, body-sentence count, and parent scenario.

2.7 Grammar-Score Generation

Grammar variants were produced from the 1,000 clean content examples. The generation plan was designed to avoid a complete lockstep relationship between grammar and content.

The 200 content-zero responses were assigned grammar score 0 because they were already generic, off-topic, or shorter than the rubric required. Each of the remaining 800 responses produced two versions:

- one lower or intermediate grammar version sampled from scores 1, 2, and 3; and
- one upper grammar version sampled from scores 4 and 5.

This resulted in 1,800 grammar-content examples. The resulting grammar distribution was approximately balanced within its planned groups:

The score-specific corruption policies were:

- **Score 5**: the clean reference was retained;
- **Score 4**: one minor orthographic error was introduced;
- **Score 3**: one genuine grammatical error was required, with an optional minor orthographic error;
- **Score 2**: at least two genuine grammatical errors were required, with an optional orthographic error;
- **Score 1**: a sentence fragment, multiple grammatical errors, and a minor orthographic error were introduced; and
- **Score 0**: the already short or non-responsive content-zero example was retained.

The perturbation engine distinguished minor orthographic errors from linguistic errors. Minor errors included missing diacritics and controlled spelling slips. Genuine grammatical errors included article or gender disagreement, number disagreement, case selection, preposition selection, incorrect preposition contractions, verb mood, verb form, missing infinitive markers, and incorrect word order. Sentence fragments were treated as the most severe operation.

Table 3: Ranges used when selecting A1–B2 anchors.

CEFR	Grammar range	Content range
A1	0–2	0–2
A2	1–3	1–3
B1	2–4	1–4
B2	3–5	2–4

The language-specific rules reflected common structural properties of each language. Examples included *de/het* selection and verb position in Dutch, article gender in Greek, case selection in Polish, gender and preposition contractions in Italian, and mood, agreement, and contraction errors in Portuguese. These rules were not intended to model every possible learner error. Their purpose was to provide controlled examples of several error families rather than relying only on random character deletion.

Each successful operation generated an error record containing:

- the error type;
- its severity;
- the original text;
- the replacement text; and
- the generation method.

Minor spelling or accent errors were assigned severity 1, grammatical errors severity 2, and fragment-producing operations severity 3. These records enabled later analysis by error category.

Identifiers, placeholders, times, percentages, and numeric codes were protected during perturbation. The generator also checked that the modified response differed from the clean response, that two inverse operations had not cancelled one another, and that protected tokens remained unchanged. Rows failing these checks were regenerated or corrected before the dataset was saved.

2.8 Synthetic CEFR Supervision

CEFR was not assigned mechanically to every grammar-content example. Doing so would have made CEFR a deterministic function of the two rubric scores. Instead, only examples considered sufficiently clear were given exact CEFR labels. The remaining rows retained their grammar and content labels but used a masked CEFR value.

The conservative ranges used to select the lower and intermediate anchors are shown in Table 3. The ranges overlap deliberately because grammar and content alone do not uniquely determine proficiency.

Within these ranges, examples were selected conservatively and balanced across languages. One hundred examples were retained for each of A1, A2, B1, and B2, corresponding to 20 examples per language at each level.

C1 and C2 could not be represented adequately by merely removing errors from lower-level responses. Advanced proficiency involves precision, lexical control, cohesion, flexible syntax, appropriate register, and the communication of qualifications or nuance. Separate C1 and C2 email variants were therefore constructed from the clean scenario content.

Both variants retained the same task facts and received grammar score 5 and content score 4. Their difference was expressed through sentence formulation, cohesion, lexical precision, and the handling of nuance. The generation process kept their sentence counts and lengths similar, preventing "the longer response is C2" from becoming a simple shortcut. Across paired C1 and C2 examples, the mean absolute length difference was approximately 2.18 words and the maximum difference was six words.

One hundred C1 and 100 C2 examples were added, again balanced across the five languages. The controlled dataset consequently contained 2,000 rows:

$$1,800 \text{ grammar-content rows} + 200 \text{ C1/C2 rows} = 2,000.$$

The exact CEFR-labelled subset contained 600 rows, with 100 examples at each level. CEFR loss was masked for the remaining controlled rows.

2.9 Authentic CEFR-Learner Data

Initial diagnostic experiments showed that the controlled CEFR labels could be predicted too easily from grammar, content, response length, and recurring expressions. To introduce more natural variation, we added 2,000 learner texts from public CEFR resources distributed through the UniversalCEFR collection.

The selected authentic subset contained:

- 645 Dutch texts from ELG CEFR-NL;
- 600 Italian texts from MERLIN;
- 446 Portuguese texts from COPLE2; and
- 309 Portuguese texts from PEAPL2.

No corresponding authentic Greek or Polish subset was incorporated in this experiment. This imbalance is a limitation of the final training resource.

These corpora primarily contain learner essays and other written responses rather than the prompt-conditioned emails used in the target task. Their prompts were unavailable or not directly compatible with the required schema. Consequently, the `questionStatement` field was left empty for these rows, and their grammar and content labels were set to the missing-label value. They contributed only to the CEFR objective.

Some COPLE2 responses appeared with adjacent CEFR labels, such as A1/A2, B1/B2, or C1/C2. Instead of duplicating the same text with two contradictory exact labels, these cases were represented as CEFR intervals. For example, a B1/B2 response was stored with a minimum level of B1 and a maximum level of B2. Exact labels were represented as intervals whose lower and upper bounds were identical.

2.10 Final Hybrid Dataset

The final training resource combined the 2,000 controlled email examples and the 2,000 authentic learner texts:

$$2,000 \text{ controlled examples} + 2,000 \text{ authentic examples} = 4,000 \text{ rows}.$$

Missing supervision was represented with the sentinel value `-100`. This allowed the loss for each task to be computed only for rows containing the corresponding annotation. Controlled examples could therefore train the grammar and content heads even when CEFR was masked, while authentic corpus examples could train the CEFR head without invented grammar or content labels.

The final dataset was divided into 3,200 training rows and 800 validation rows. For controlled data, the previously defined scenario-level division was retained. Authentic texts were split by their source grouping or text identity to prevent identical responses from appearing in both partitions.

This hybrid design provided substantially broader CEFR supervision, but it did not remove every source mismatch. The controlled portion contains structured emails and programmed perturbations, while the authentic portion contains naturally occurring learner writing from different domains. The consequences of this mismatch are examined in the results and limitations sections.

3 Preprocessing and Input Representation

3.1 Text Normalization

All prompts and responses were stored as Unicode strings and normalized to Unicode NFC form. This was particularly important for Greek, Polish, Portuguese, and Italian, where visually identical accented characters can have different underlying Unicode representations.

The text was not lowercased and diacritics were not removed. Capital letters, accents, cedillas, and other language-specific marks can contain useful grammatical and orthographic information. For example, removing these distinctions would make it impossible for the model to distinguish a correctly written Portuguese word from a generated missing-diacritic error.

Whitespace between email paragraphs was retained. No stemming, lemmatization, stop-word removal, or external grammatical correction was applied. The purpose was to allow the pretrained encoder to observe the response in approximately the same form in which it would be written by a candidate.

3.2 Label Representation

Grammar and content were represented as numerical labels:

$$y^{(g)} \in \{0,1,2,3,4,5\}, \qquad y^{(c)} \in \{0,1,2,3,4\}.$$

CEFR levels were mapped to ordered integer identifiers:

$$\text{A1} = 0, \quad \text{A2} = 1, \quad \text{B1} = 2, \quad \text{B2} = 3, \quad \text{C1} = 4, \quad \text{C2} = 5.$$

Three CEFR fields were retained:

- an exact CEFR label, when one was available;
- a minimum CEFR label; and
- a maximum CEFR label.

For an exact B1 example, all three values were 2. For an interval example labelled B1/B2, the exact label was unavailable, the minimum was 2, and the maximum was 3.

The value `-100` represented missing supervision. It was used for grammar and content labels in the authentic learner corpora and for CEFR labels in controlled examples that were not selected as reliable anchors. The missing value was not treated as an additional class. It was removed through task-specific masks before each loss was calculated.

3.3 Prompt and Response as a Text Pair

Both the writing prompt and the candidate response are necessary for the task. Grammar can be estimated largely from the response, but content coverage cannot be assessed without knowing what the prompt requested. CEFR may also depend on whether the writer uses language appropriate to the communicative situation.

The prompt and response were therefore passed to the tokenizer as a text pair:

```
tokenizer(questionStatement, text).
```
(1)

For XLM-RoBERTa, this produces a sequence with the model's special separators, conceptually of the form

$$\langle s\rangle \text{ prompt } \langle /s\rangle\langle /s\rangle \text{ email } \langle /s\rangle.$$

The two inputs were not concatenated manually. Pair encoding allowed the tokenizer to insert the separator tokens expected by the pretrained model.

No explicit language identifier was provided to the encoder. The model was expected to infer the language from its multilingual subword representations. This also avoided making language a direct shortcut for any target score.

Authentic learner-corpus rows did not have compatible email prompts. For these examples, the prompt field was an empty string and the learner text occupied the response field. These rows could supervise CEFR but were masked for grammar and content.

3.4 Subword Tokenization

We used the pretrained tokenizer distributed with `xlm-roberta-base`. It applies the multilingual SentencePiece vocabulary learned with the original XLM-RoBERTa model. Training a new tokenizer was avoided because the pretrained encoder's embedding matrix is tied to its existing vocabulary. Replacing it would discard a substantial part of the model's multilingual pretraining.

Before truncation, the hybrid dataset had the following token-length distribution:

- mean length: approximately 200 tokens;
- median length: 174 tokens;
- 90th percentile: 344 tokens;
- 95th percentile: 368 tokens;
- 99th percentile: approximately 413 tokens; and
- maximum length: 552 tokens.

The longer examples were mainly authentic learner texts rather than the controlled emails. A maximum input length of 384 subword tokens was used during the hybrid-data experiments. This retained the full sequence for most examples while keeping training feasible on a single NVIDIA T4 GPU. Sequences longer than this value were truncated using the tokenizer's pair-truncation behaviour.

This choice introduces a limitation. A small fraction of the longest authentic texts were not represented in full. Increasing the limit to 512 would preserve more text but would also increase attention-memory and computation costs. Since the target application consists of relatively short emails, 384 tokens was considered a reasonable trade-off.

3.5 Dynamic Padding

Examples were tokenized with

```
padding=False.
```

This did not mean that padding was omitted from model batches. It meant that padding was postponed until batch construction. Storing every example at the global maximum length would add hundreds of unnecessary padding tokens to many short emails.

A custom data collator collected the examples in each minibatch and padded them to the length of the longest sequence in that batch. The padded length was rounded to a multiple of eight:

$$L_{\text{batch}} = 8 \left\lceil \frac{\max_i L_i}{8} \right\rceil.$$

Padding to a multiple of eight can improve the efficiency of mixed-precision operations on compatible GPUs. More importantly, dynamic padding reduced computation compared with padding every example to 384 tokens.

The collator also converted the labels into tensors. Grammar and content labels used floating-point tensors because they were trained as regression targets. CEFR exact, minimum, and maximum labels used integer tensors because they defined ordinal classes and intervals.

3.6 Attention Masks

Padding tokens should not contribute to the contextual representation. The tokenizer therefore generated an attention mask for every input:

$$m_{i,t} = \begin{cases} 1, & \text{if position } t \text{ contains a real token,} \\ 0, & \text{if position } t \text{ contains padding.} \end{cases}$$

Both `input_ids` and `attention_mask` were passed to XLM-RoBERTa. The input identifiers specify which subword occurs at each position, while the attention mask tells the model which positions are valid.

The same mask was later used for mean pooling. If $\mathbf{h}_{i,t}$ denotes the final hidden representation at token position t, the masked mean representation is

$$\mathbf{h}_i = \frac{\sum_t m_{i,t}\mathbf{h}_{i,t}}{\max\left(1, \sum_t m_{i,t}\right)}. \tag{2}$$

Consequently, padding tokens did not influence the pooled email representation.

3.7 Task-Specific Label Masks

Every minibatch could contain examples with different forms of supervision. Three masks were constructed:

$$M_g = I\left[y^{(g)} \neq -100\right], \tag{3}$$

$$M_c = I\left[y^{(c)} \neq -100\right], \tag{4}$$

$$M_e = I\left[y^{(\min)} \neq -100 \wedge y^{(\max)} \neq -100\right]. \tag{5}$$

The grammar loss was calculated only over M_g, the content loss only over M_c, and the CEFR loss only over M_e. This was the mechanism that allowed controlled and authentic data to coexist in a single minibatch.

An authentic learner-text row could update the shared encoder, the shared projection, and the CEFR head without contributing to the grammar or content losses. A controlled email with masked CEFR could update the grammar and content heads without being forced into an uncertain proficiency class.

3.8 Preprocessing Validation

Before training, the processed datasets were checked to ensure that:

- all required model inputs were present;
- grammar and content labels had the expected numerical ranges;
- exact CEFR labels were between 0 and 5;
- CEFR interval bounds were valid and ordered;
- missing labels were consistently represented by `-100`;
- train and validation rows remained separated according to the stored split;
- tokenized batches contained matching input and attention-mask shapes; and
- the collator preserved all five label tensors.

After collation, a batch had the following fields:

```
input_ids
attention_mask
grammar_labels
content_labels
cefr_labels
cefr_min_labels
cefr_max_labels
```

The raw prompt and response columns were removed from the tokenized model dataset after encoding. This reduced duplication in memory. The original data files remained unchanged and were retained for auditing, analysis, and inference.

4 Model Architecture

4.1 Multitask Formulation

The model receives a tokenized writing prompt and candidate response and produces three outputs:

$$\hat{y}^{(g)} \in [0, 5], \tag{6}$$

$$\hat{y}^{(c)} \in [0, 4], \tag{7}$$

$$\hat{y}^{(e)} \in \{0, 1, 2, 3, 4, 5\}. \tag{8}$$

The first two outputs are continuous predictions for grammar and content. The third is an ordinal CEFR prediction, where the integer values correspond to A1 through C2.

The shared multilingual encoder was used because the three tasks depend on overlapping linguistic evidence. Grammatical errors, missing task details, vocabulary, cohesion, and sentence structure are all present in the same response. At the same time, separate prediction heads were retained because the three labels represent different assessment criteria.

The architecture consists of four main stages:

1. an XLM-RoBERTa multilingual encoder;
2. a sequence-pooling operation;
3. a shared nonlinear projection; and
4. three task-specific output heads.

Conceptually, the computation is

$$(\text{prompt}, \text{email}) \longrightarrow \text{XLM-R} \longrightarrow \text{pooling} \longrightarrow \text{shared projection} \longrightarrow \begin{cases} \text{grammar head}, \\ \text{content head}, \\ \text{ordinal CEFR head}. \end{cases} \tag{9}$$

4.2 Multilingual Encoder

We used `xlm-roberta-base` as the pretrained encoder. XLM-R provides a shared multilingual representation space and supports all five target languages through a common SentencePiece vocabulary. The base model contains 12 Transformer layers, a hidden size of 768, and 12 self-attention heads per layer.

Given input identifiers $\mathbf{x}$ and an attention mask $\mathbf{m}$, the encoder produces a final hidden representation for every token:

$$\mathbf{H} = \mathrm{XLMR}(\mathbf{x}, \mathbf{m}), \qquad \mathbf{H} \in R^{T \times 768}, \tag{10}$$

where T is the padded sequence length. Each row $\mathbf{h}_t$ contains the contextual representation of the token at position t. Since the prompt and response are encoded together, these representations can capture relationships between task requirements and the submitted email, as discussed above in the previous section.

4.3 Parameter-Efficient Adaptation

Fully fine-tuning XLM-RoBERTa would update approximately 278 million parameters. This was unnecessary for the available dataset and would increase memory usage and the risk of overfitting and no availability of GPU. We therefore used low-rank adaptation, or LoRA, while keeping the original encoder weights frozen.

For a frozen weight matrix $\mathbf{W}$, LoRA represents the learned update as

$$\mathbf{W}'\mathbf{x} = \mathbf{W}\mathbf{x} + \frac{\alpha}{r}\mathbf{B}\mathbf{A}\mathbf{x}, \tag{11}$$

where

$$\mathbf{A} \in R^{r \times d_{\text{in}}}, \qquad \mathbf{B} \in R^{d_{\text{out}} \times r}.$$

The update $\mathbf{BA}$ has rank at most r, making it much smaller than a full $d_{\text{out}} \times d_{\text{in}}$ weight update.

The LoRA configuration was:

- rank $r = 8$;
- scaling parameter $\alpha = 16$;
- LoRA dropout = 0.10;
- target modules: attention query and value projections; and
- no trainable bias terms inside the adapters.

The query projections influence which token relationships receive attention, while the value projections determine what information is transmitted through those relationships. Adapting these two projections is a commonly used parameter-efficient compromise. Key projections and feed-forward layers remained frozen.

Across the 12 encoder layers, the query and value adapters contained 294,912 trainable parameters. The shared projection and task-specific heads added approximately 199,000 parameters. Thus, the complete model trained roughly 494,000 parameters, which is about 0.18% of the overall model size.

4.4 Sequence Pooling

XLM-RoBERTa produces one vector per input token, but the prediction heads require one vector for the complete prompt-response pair. Two pooling strategies were evaluated.

4.4.1 First-Token Pooling

The first experiment used the final representation of the initial $\langle s \rangle$ token:

$$\mathbf{h}_{\text{first}} = \mathbf{H}_0. \tag{12}$$

This token occupies the first position in every XLM-R input and can accumulate information from the complete sequence through self-attention. This experiment is referred to as the first-token or CLS-style pooling model.

Although XLM-R uses $\langle s \rangle$ rather than a token literally named `[CLS]`, the term CLS-style pooling describes the same operation of selecting the first contextual representation.

4.4.2 Attention-Mask-Aware Mean Pooling

A second experiment used the mean of all non-padding token representations:

$$\mathbf{h}_{\text{mean}} = \frac{\sum_{t=1}^{T} m_t \mathbf{h}_t}{\max\left(1, \sum_{t=1}^{T} m_t\right)}. \tag{13}$$

The attention mask excludes padded positions. Special tokens remain part of the average because their attention-mask values are one.

Mean pooling was tested because writing-quality evidence is distributed across the response. A grammatical error may occur near the end of one sentence, while a required task detail may occur in a different paragraph. Averaging the contextual token representations gives every non-padding position a direct contribution to the pooled vector.

The pooling comparison was performed as a controlled experiment. The training data, random seed, LoRA configuration, optimizer, losses, and number of epochs were kept unchanged. Only the pooling operation was modified, and the model was retrained from its pretrained initialization.

4.5 Shared Projection Layer

The pooled 768-dimensional vector was passed through a shared projection:

$$\mathbf{z} = \mathrm{Dropout}\left(\mathrm{GELU}\left(\mathbf{W}_p\,\mathrm{LayerNorm}(\mathbf{h}) + \mathbf{b}_p\right)\right), \tag{14}$$

where

$$\mathbf{W}_p \in R^{256 \times 768}, \qquad \mathbf{z} \in R^{256}.$$

The projection consisted of:

1. layer normalization;
2. a linear transformation from 768 to 256 dimensions;

3. a GELU activation; and

4. dropout with probability 0.10.

Layer normalization stabilized the scale of the pooled representation. The projection reduced the dimensionality before the prediction heads and provided a shared trainable transformation for all three tasks. The dropout layer acted as regularization during training and was disabled during evaluation and inference.

4.6 Grammar Regression Head

The grammar head was a linear layer producing one scalar logit:

$$a_g = \mathbf{w}_g^\top \mathbf{z} + b_g. \tag{15}$$

The logit was transformed into the rubric range using a sigmoid:

$$\hat{y}^{(g)} = 5\sigma(a_g). \tag{16}$$

Since $\sigma(a_g) \in (0, 1)$, the final prediction remains between 0 and 5. The output was kept continuous during training and evaluation. It was not rounded before computing Pearson correlation.

The sigmoid was used as a range constraint rather than as a binary classifier. It prevented the regression head from producing values outside the valid grammar scale.

4.7 Content Regression Head

The content head followed the same design:

$$a_c = \mathbf{w}_c^\top \mathbf{z} + b_c, \tag{17}$$

$$\hat{y}^{(c)} = 4\sigma(a_c). \tag{18}$$

Multiplication by four restricted the prediction to the content rubric range $[0, 4]$. The grammar and content heads had independent parameters, allowing the same shared representation to support different scoring criteria.

4.8 Ordinal CEFR Head

CEFR classes have a natural order:

$$\text{A1} < \text{A2} < \text{B1} < \text{B2} < \text{C1} < \text{C2}.$$

A standard six-class softmax treats these levels as unrelated categories. Predicting A1 instead of A2 would be penalized in the same categorical manner as predicting A1 instead of C2. We therefore modelled CEFR as an ordinal problem.

The CEFR head first computes a scalar proficiency score:

$$s = \mathbf{w}_e^\top \mathbf{z} + b_e. \tag{19}$$

Five learned thresholds divide this continuous axis into six ordered regions:

$$\theta_1 < \theta_2 < \theta_3 < \theta_4 < \theta_5.$$

Table 4: Ordinal CEFR target representation.

CEFR level	Ordinal target
A1	$[0, 0, 0, 0, 0]$
A2	$[1, 0, 0, 0, 0]$
B1	$[1, 1, 0, 0, 0]$
B2	$[1, 1, 1, 0, 0]$
C1	$[1, 1, 1, 1, 0]$
C2	$[1, 1, 1, 1, 1]$

Each threshold answers one cumulative question:

$$k = 0 : P(y > \text{A1}), \tag{20}$$

$$k = 1 : P(y > \text{A2}), \tag{21}$$

$$k = 2 : P(y > \text{B1}), \tag{22}$$

$$k = 3 : P(y > \text{B2}), \tag{23}$$

$$k = 4 : P(y > \text{C1}). \tag{24}$$

The corresponding logits are

$$\ell_k = s - \theta_k, \qquad k \in \{1, \ldots, 5\}. \tag{25}$$

The cumulative probabilities are

$$p_k = \sigma(\ell_k) = P(y > \text{level}_k). \tag{26}$$

For example, a B2 target should pass the first three thresholds but not the final two. Its ordinal target is therefore

$$[1, 1, 1, 0, 0].$$

The complete target encoding is shown in Table 4.

4.9 Ordered Threshold Parameterization

The threshold order was enforced by parameterization rather than by a post-update sorting operation. The first threshold was represented by one trainable scalar θ_1. Four additional unconstrained parameters $r_2, \ldots, r_5$ defined positive steps:

$$\Delta_k = \text{softplus}(r_k) > 0. \tag{27}$$

The remaining thresholds were constructed cumulatively:

$$\theta_2 = \theta_1 + \Delta_2, \tag{28}$$

$$\theta_3 = \theta_2 + \Delta_3, \tag{29}$$

$$\theta_4 = \theta_3 + \Delta_4, \tag{30}$$

$$\theta_5 = \theta_4 + \Delta_5. \tag{31}$$

Because every softplus output is strictly positive, a later threshold must always be greater than the previous one. Gradient updates may change the threshold spacing, but they cannot reverse the order.

The initial threshold values were

$$[-2, -1, 0, 1, 2].$$

To obtain unit initial gaps, each raw step was initialized as

$$r_k = \log(e^1 - 1), \tag{32}$$

because

$$\text{softplus}\,(\log(e-1)) = 1.$$

This produced five evenly spaced initial decision boundaries while allowing their positions and separations to be learned.

4.10 CEFR Prediction

The uncalibrated hard CEFR prediction was the number of thresholds whose probability exceeded 0.5:

$$\hat{y}^{(e)}_{\text{hard}} = \sum_{k=1}^{5} I[p_k > 0.5]. \tag{33}$$

The result is an integer from 0 to 5 and maps directly to A1 through C2. For example, passing three thresholds gives class 3, corresponding to B2.

A continuous ordinal score was also calculated:

$$\hat{y}^{(e)}_{\text{soft}} = \sum_{k=1}^{5} p_k. \tag{34}$$

This value also lies between 0 and 5, but it retains the model's uncertainty near class boundaries. It was used for diagnostic Pearson correlation and for post-training calibration. It was not itself an additional trainable output head.

4.11 Model Variants Used in the Experiments

Four main experiments were conducted. The project directory used incremental version numbers during development, but descriptive names are used in this report to make the progression clearer.

4.11.1 Synthetic-5E Baseline

The first complete experiment used the 2,000-row controlled dataset, first-token pooling, and five training epochs. This model established that the complete pipeline could jointly learn grammar, content, and ordinal CEFR predictions. It also provided the initial reference point for later experiments.

This experiment is stored under Version 2 of the project. Version 2 also contains the main controlled-data construction pipeline, including the scenario bank, content variants, grammar perturbations, and synthetic CEFR anchors.

4.11.2 Synthetic-8E Experiment

The second experiment used the same controlled dataset and the same model architecture but increased the training schedule from five to eight epochs. The model was trained again from the pretrained XLM-RoBERTa initialization rather than continuing from the five-epoch checkpoint.

This experiment was designed to distinguish limitations caused by insufficient optimization from limitations caused by the training data. It is stored under Version 3 of the project.

4.11.3 Hybrid-CLS Experiment

The third experiment augmented the 2,000 controlled examples with 2,000 authentic CEFR-labelled learner texts. The resulting hybrid dataset contained 4,000 rows.

This required two changes to the learning procedure. First, task losses were masked independently because the authentic texts did not have grammar or content labels. Second, the CEFR objective was extended to support both exact labels and adjacent-level intervals.

The encoder, LoRA configuration, shared projection, prediction heads, and first-token pooling operation were otherwise retained. The model was trained for eight epochs. This experiment is stored under Version 4 of the project.

4.11.4 Hybrid-Mean Experiment

The fourth experiment used the same hybrid dataset, label masks, ordinal objectives, LoRA configuration, optimizer settings, and eight-epoch schedule as the Hybrid-CLS experiment. The intended architectural change was limited to the sequence-pooling operation.

The representation of the first $\langle s \rangle$ token was replaced with an attention-mask-aware average of the non-padding token representations. The purpose was to test whether distributed evidence, such as grammatical errors and task details appearing throughout the email, could be captured more effectively through mean pooling.

This experiment is stored under Version 5 of the project.

4.11.5 Experimental Progression

The four experiments form a controlled progression:

$$\begin{aligned} \text{Synthetic-5E} &\longrightarrow \text{Synthetic-8E} \\ &\longrightarrow \text{Hybrid-CLS} \\ &\longrightarrow \text{Hybrid-Mean}. \end{aligned} \tag{35}$$

The first comparison studies training duration. The second studies the effect of adding authentic CEFR supervision. The third studies the effect of the pooling strategy. This progression allows the influence of each major design change to be discussed separately rather than treating the final model as a single unexplained configuration.

5 Training Objectives

5.1 Masked Multitask Learning

The model was trained jointly on grammar, content, and CEFR. However, not every row contained labels for all three tasks. Controlled emails generally contained grammar and content labels, while authentic learner texts contained only CEFR supervision. Some controlled

Table 5: Summary of the four principal model experiments.

Experiment name	Project version	Training data	Pooling	Epochs	Purpose
Synthetic-5E	Version 2	2,000 controlled email examples	First token	5	Establish the first complete multitask baseline.
Synthetic-8E	Version 3	Same 2,000 controlled email examples	First token	8	Measure the effect of longer training without changing the data or architecture.
Hybrid-CLS	Version 4	2,000 controlled emails and 2,000 authentic learner texts	First token	8	Measure whether authentic CEFR supervision improves generalization beyond the controlled data.
Hybrid-Mean	Version 5	Same 4,000-example hybrid dataset	Masked mean	8	Isolate the effect of replacing the first-token representation with attention-mask-aware mean pooling.

emails also had their CEFR labels masked because their proficiency level was considered ambiguous.

A separate loss was therefore computed for each task using only the rows for which that task's label was available. If a minibatch contained no valid labels for a particular task, that loss term was omitted for the minibatch.

The total loss was

$$\mathcal{L} = \mathcal{L}_{\text{grammar}} + \mathcal{L}_{\text{content}} + \mathcal{L}_{\text{CEFR}}, \tag{36}$$

where only the available terms were included. No manually selected task weights were applied in the principal experiments.

This formulation allowed examples with different annotations to update the same encoder. A CEFR-only learner text could improve the shared multilingual representation and the CEFR head, while a controlled email with masked CEFR could still update the grammar and content heads.

5.2 Grammar Objective

Grammar was treated as bounded regression rather than six-class classification. This reflects the ordered nature of the rubric. A prediction of 3.8 for a response labelled 4 should be considered closer than a prediction of 1.0.

The model prediction was

$$\hat{y}^{(g)} = 5\sigma(a_g), \tag{37}$$

which constrained it to the valid interval $[0, 5]$.

Before computing the loss, both prediction and target were divided by five:

$$\mathcal{L}_{\text{grammar}} = \frac{1}{N_g} \sum_{i \in \mathcal{I}_g} \left(\frac{\hat{y}_i^{(g)}}{5} - \frac{y_i^{(g)}}{5} \right)^2, \tag{38}$$

where

$$\mathcal{I}_g = \left\{ i : y_i^{(g)} \neq -100 \right\}$$

is the set of rows with grammar supervision, and $N_g = |\mathcal{I}_g|$.

Dividing by five maps both values to $[0, 1]$. Without this normalization, grammar errors measured on a six-point scale would produce larger squared losses than content errors measured on a five-point scale. The normalization therefore makes the magnitudes of the two regression objectives more comparable.

5.3 Content Objective

Content was also treated as bounded regression. The model prediction was

$$\hat{y}^{(c)} = 4\sigma(a_c), \tag{39}$$

which lies in $[0, 4]$.

The normalized content loss was

$$\mathcal{L}_{\text{content}} = \frac{1}{N_c} \sum_{i \in \mathcal{I}_c} \left(\frac{\hat{y}_i^{(c)}}{4} - \frac{y_i^{(c)}}{4} \right)^2, \tag{40}$$

where

$$\mathcal{I}_c = \left\{ i : y_i^{(c)} \neq -100 \right\}.$$

The content prediction remained continuous during training and evaluation. Rounding was unnecessary for Pearson correlation and would discard useful information about the model's confidence between adjacent scores.

5.4 Exact Ordinal CEFR Targets

For an exact CEFR label $y_i \in \{0, \ldots, 5\}$, five cumulative binary targets were created:

$$t_{i,k} = I[y_i > k], \qquad k \in \{0, 1, 2, 3, 4\}. \tag{41}$$

For example, B2 corresponds to $y_i = 3$, producing

$$\mathbf{t}_i = [1, 1, 1, 0, 0].$$

The model generated five logits

$$\ell_{i,k} = s_i - \theta_k. \tag{42}$$

Binary cross-entropy with logits was applied independently to the five ordinal decisions:

$$\mathcal{L}_{\text{CEFR}}^{(i)} = \frac{1}{5} \sum_{k=0}^{4} \text{BCEWithLogits}\left(\ell_{i,k}, t_{i,k}\right). \tag{43}$$

The implementation used binary cross-entropy directly on the logits rather than first applying a sigmoid. The combined `BCEWithLogits` operation is numerically more stable than computing probabilities and logarithms separately.

5.5 Interval CEFR Supervision

Some authentic responses had adjacent-level labels, such as A1/A2, B1/B2, or C1/C2. Forcing one endpoint to be the exact label would add false certainty. Instead, the response was represented by an interval

$$[l_i, u_i], \qquad 0 \le l_i \le u_i \le 5.$$

An exact label is a special case where $l_i = u_i$.

For every CEFR threshold k, three cases were considered:

$$t_{i,k} = \begin{cases} 1, & k < l_i, \\ 0, & k \ge u_i, \\ \text{unknown}, & l_i \le k < u_i. \end{cases} \tag{44}$$

The interpretation is as follows:

- if $k < l_i$, the response is certainly above threshold k;
- if $k \ge u_i$, the response is certainly not above threshold k; and
- if the threshold lies inside the interval, the available annotation does not determine which side is correct.

Consider a B1/B2 response with

$$l_i = 2, \qquad u_i = 3.$$

Its threshold supervision becomes

$$[1, 1, ?, 0, 0],$$

where ? denotes an ignored target. The model is taught that the response is above A1 and A2 and below C1 and C2, but it is not forced to decide whether it lies above the B1 boundary.

A binary mask $q_{i,k}$ identified the known threshold targets:

$$q_{i,k} = I[k < l_i] + I[k \ge u_i]. \tag{45}$$

The interval-aware CEFR loss was

$$\mathcal{L}_{\text{CEFR}} = \frac{\sum_{i,k} q_{i,k} \, \text{BCEWithLogits}\left(\ell_{i,k}, t_{i,k}\right)}{\sum_{i,k} q_{i,k}}. \tag{46}$$

Rows with missing CEFR bounds were excluded completely. Thresholds inside a valid uncertainty interval were also excluded, while all thresholds known to lie outside the interval remained supervised.

5.6 Why an Ordinal Objective Was Used

A six-class softmax could predict CEFR, but it would not explicitly encode the relationship between adjacent levels. In a standard categorical objective, the class indices have no mathematical order. The labels A1 and A2 are simply two categories, just as A1 and C2 are two categories.

The cumulative ordinal objective incorporates the order directly. Moving from A1 to C2 requires crossing all five thresholds. Adjacent errors normally differ by one threshold, while distant errors differ across several thresholds.

This representation also supports interval supervision naturally. An uncertain adjacent boundary can be masked while retaining information about the remaining boundaries. Achieving the same behaviour with a standard six-class target would require either duplicating the example with conflicting labels or assigning an arbitrary probability distribution.

5.7 Loss Scaling and Task Interaction

The normalized grammar and content losses lie on approximately comparable scales. The CEFR loss is averaged over its valid ordinal decisions. The three available losses were then summed with equal weight.

This equal weighting was chosen as a simple baseline. It avoids tuning task weights on a small validation set and makes the training objective easy to reproduce. However, equal numerical weights do not guarantee equal influence on the shared encoder. The tasks differ in sample count, difficulty, label noise, and gradient magnitude.

The multitask setting can provide useful inductive transfer. For example, representations that detect omitted information may assist content scoring, while representations sensitive to agreement and word order may assist grammar and CEFR. It can also create interference if one task encourages shortcuts that do not generalize to the others. This possibility motivated the separate reporting of all three validation metrics and the later analysis of out-of-distribution performance.

5.8 Calibration Was Not Part of Training

CEFR calibration was performed only after model training. It replaced the fixed conversion from the continuous CEFR score to discrete levels with cut-points fitted on validation predictions.

Calibration did not:

- change the XLM-RoBERTa parameters;
- update the LoRA adapters;
- modify the shared projection;
- modify the prediction heads; or
- affect the training loss.

It changed only the final mapping from a continuous ordinal score to one of the six CEFR labels. The calibration procedure is described separately in Section 7.

6 Training Configuration

6.1 Training and Validation Partitions

The final hybrid dataset contained 4,000 examples. It was divided into 3,200 training rows and 800 validation rows.

For the controlled email data, the split was defined at the scenario level. All language versions, content variants, grammar variants, and CEFR variants derived from one scenario remained in the same partition. This prevented localized or perturbed versions of the same underlying task from appearing in both training and validation.

The authentic learner texts retained their source and text identifiers during splitting. Duplicate text instances and adjacent-label versions of the same response were consolidated before the final partition was created.

Training batches were shuffled at the beginning of every epoch. Validation batches were not shuffled.

6.2 Hardware and Software Setting

The experiments were run in Google Colab using a single NVIDIA T4 GPU. Training used PyTorch, Hugging Face Transformers, PEFT, and the Hugging Face Datasets library.

The encoder was loaded from the `xlm-roberta-base` checkpoint. Only the LoRA parameters, shared projection, prediction heads, and ordinal thresholds were optimized. The original XLM-RoBERTa parameters remained frozen.

A fixed seed of 42 was applied to Python, NumPy, PyTorch, and all available CUDA devices:

```
random.seed(42)
numpy.random.seed(42)
torch.manual_seed(42)
torch.cuda.manual_seed_all(42)
```

These settings improve reproducibility, although exact bitwise reproduction is not guaranteed across different GPU types, CUDA versions, and library versions.

6.3 Batching and Gradient Accumulation

The training dataloader used a minibatch size of eight examples. Dynamic padding was performed independently for each batch. Two minibatches were accumulated before each optimizer update, giving an effective batch size of

$$B_{\text{effective}} = 8 \times 2 = 16. \tag{47}$$

If the final minibatch of an epoch did not complete the accumulation interval, an optimizer update was still performed. Validation used larger inference-only batches where memory permitted because gradient storage was unnecessary.

For each minibatch, the loss was divided by the number of accumulation steps before backpropagation:

$$\widetilde{\mathcal{L}} = \frac{\mathcal{L}}{N_{\text{accum}}}. \tag{48}$$

Table 6: Optimizer parameter groups.

Parameter group	Learning rate	Weight decay
LoRA adapters	2×10^{-4}	0.01
Projection and heads	5×10^{-4}	0.01

This ensures that the accumulated gradient approximates the gradient of a single larger batch rather than doubling its magnitude.

6.4 Optimizer

AdamW was used with two parameter groups. The LoRA encoder parameters and the newly initialized task components used different learning rates:

The shared projection and prediction heads were initialized for the new task, so they were assigned a larger learning rate. The LoRA parameters adapt representations that already originate from a pretrained multilingual model and therefore used a smaller learning rate.

The remaining AdamW settings used the PyTorch defaults. The frozen XLM-RoBERTa parameters were not included in either optimizer group.

6.5 Learning-Rate Schedule

A linear schedule with warm-up was used. Let U denote the total number of optimizer updates across all epochs. The number of warm-up updates was

$$U_{\text{warmup}} = \lfloor 0.10U \rfloor. \tag{49}$$

During the first 10% of optimizer updates, the learning rate increased linearly from zero to its specified value. It then decreased linearly to zero over the remaining updates.

The scheduler advanced after every optimizer update, not after every minibatch. This distinction is necessary because gradient accumulation means that several minibatches can correspond to one optimizer step.

For the hybrid experiments, 3,200 training rows and a minibatch size of eight produced 400 minibatches per epoch. With two-step gradient accumulation, this corresponded to approximately 200 optimizer updates per epoch.

6.6 Mixed-Precision Training

Training used automatic mixed precision on the GPU. The forward pass and loss calculation were executed inside a CUDA autocast context using 16-bit floating-point operations where supported.

A gradient scaler was used to reduce numerical underflow:

1. scale the loss;
2. perform backpropagation on the scaled loss;
3. unscale the gradients before clipping;
4. apply the optimizer update; and

5. update the scaling factor.

The model parameters themselves were not manually converted entirely to 16-bit values. Automatic mixed precision selected suitable operations while retaining higher precision where needed.

6.7 Gradient Clipping and Resetting

Before every optimizer update, gradients were unscaled and clipped to a maximum global norm of 1.0:

$$\|\mathbf{g}\|_2 \leq 1. \tag{50}$$

Gradient clipping was included to reduce the effect of occasional large updates produced by mixed batches of exact, interval, and multitask supervision.

Gradients were cleared using

```
optimizer.zero_grad(set_to_none=True)
```

before the first training iteration and after every optimizer update. They were not reset at the start of every minibatch because gradient accumulation intentionally combines gradients from two minibatches. The LoRA matrices and prediction-head parameters were initialized when the model was created. They did not require a separate manual "set-to-zero" operation before each epoch.

6.8 Number of Epochs

The first controlled-data model was trained for five epochs. A second controlled-data experiment was trained from the pretrained initialization for eight epochs.

The two hybrid-data models were each trained for eight epochs:

- Hybrid-CLS used first-token pooling;
- Hybrid-Mean used attention-mask-aware mean pooling.

The eight-epoch models were trained from fresh pretrained XLM-RoBERTa initializations. They were not obtained by continuing the five-epoch checkpoint for three additional epochs. This made the training schedules independently reproducible.

6.9 Validation After Each Epoch

After every training epoch, the model was switched to evaluation mode and applied to the complete validation dataloader without gradient calculation.

The following quantities were recorded:

- mean validation loss;
- grammar Pearson correlation;
- content Pearson correlation;
- CEFR Pearson correlation using hard predictions;
- CEFR Pearson correlation using the continuous ordinal score;
- CEFR exact accuracy;
- mean training loss; and
- a composite checkpoint-selection score.

Grammar and content metrics were calculated only over examples with the corresponding labels. Exact CEFR accuracy was calculated only over rows with exact CEFR labels. Interval-labelled rows contributed to training but were excluded from exact-label accuracy.

6.10 Checkpoint-Selection Criterion

The checkpoint-selection score was the unweighted mean of four validation metrics:

$$S_{\text{select}} = \frac{1}{4}\left(r_g + r_c + r_{e,\text{soft}} + a_e\right), \tag{51}$$

where

- r_g is grammar Pearson correlation;
- r_c is content Pearson correlation;
- $r_{e,\text{soft}}$ is CEFR Pearson correlation using the continuous ordinal score; and
- a_e is exact CEFR accuracy.

The continuous CEFR correlation was used for checkpoint selection because it preserves information near class boundaries. Exact accuracy was included to ensure that improvement in the continuous ordering did not come at the expense of the required discrete prediction.

Whenever the selection score improved, copies of all trainable parameters were stored in CPU memory. This included the LoRA adapters, shared projection, grammar head, content head, CEFR score layer, and learned CEFR thresholds.

This was a form of validation-based model selection rather than training on the validation data. Gradients were never computed from validation examples.

6.11 Checkpoint Serialization

The final checkpoint was saved in separate components:

- the LoRA adapter in Hugging Face PEFT format;
- the tokenizer files;
- the shared projection and prediction-head state dictionaries;
- the ordinal CEFR threshold parameters; and
- the CEFR calibration cut-points in a JSON file.

The LoRA weights were stored in a `model.safetensors` file together with the adapter configuration. The frozen XLM-RoBERTa base weights were not duplicated inside the submission checkpoint. During reloading, the base model is instantiated first, the adapter is attached, and the separately saved projection and prediction heads are restored.

The calibration JSON file is loaded only for the final CEFR discretization step. It is not part of the model's trainable state.

Table 7: Principal training hyperparameters.

Setting	Value
Base encoder	XLM-RoBERTa-base
Maximum length	384 tokens
Training batch size	8
Gradient accumulation	2 steps
Effective batch size	16
Training epochs	8
LoRA rank	8
LoRA alpha	16
LoRA dropout	0.10
Projection size	256
Projection dropout	0.10
LoRA learning rate	2×10^{-4}
Head learning rate	5×10^{-4}
Weight decay	0.01
Warm-up proportion	0.10
Scheduler	Linear decay
Gradient clipping	1.0
Numerical precision	Mixed FP16
Optimizer	AdamW
Random seed	42

6.12 Training Hyperparameters

The principal hybrid-experiment configuration is summarized in Table 7.

No large hyperparameter search was performed. The experimental effort focused on the data composition, training duration, CEFR supervision, and pooling strategy. This limited search should(might) be considered when interpreting the final results.

7 Evaluation Methodology

7.1 Evaluation Partition

Model development and selection used the internal validation partition created during dataset construction.

For the controlled portion of the dataset, the partition was defined at the scenario level. No localized, content-modified, or grammatically perturbed version of a validation scenario appeared in the training partition. This prevented the model from being evaluated on alternative versions of an underlying task it had already observed.

For the authentic learner-text portion, rows were separated using their text identities and source information. Duplicate texts and multiple label records associated with the same response were kept within a single partition.

The validation partition was used for:

- evaluation after every training epoch;
- selection of the best model checkpoint;
- comparison of the four principal experiments; and
- fitting the post-training CEFR calibration cutpoints.

All validation metrics were computed after switching the model to evaluation mode. Gradient calculation was disabled, and validation examples were never used for optimizer updates.

Because the same validation partition was used for checkpoint selection and CEFR calibration, calibrated validation accuracy should be interpreted as a development result rather than as an unbiased estimate of deployment performance. Both calibrated and uncalibrated metrics were therefore retained for analysis.

7.2 Pearson Correlation

Pearson correlation was the principal metric for grammar, content, and ordinal CEFR predictions. For targets $\mathbf{y}$ and predictions $\hat{\mathbf{y}}$, it is defined as

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2}\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}. \tag{52}$$

Pearson correlation measures whether larger target scores are associated with larger predictions. It does not require predictions to be exact integers and is therefore suitable for the continuous grammar and content outputs.

A safe implementation returned zero when:

- fewer than two valid examples were available;
- all valid target values were identical; or
- all predictions were identical.

This prevented undefined correlations from propagating through the checkpoint-selection procedure.

Grammar Pearson correlation was computed only over rows satisfying

$$y_i^{(g)} \neq -100,$$

and content correlation was computed only over rows satisfying

$$y_i^{(c)} \neq -100.$$

The continuous predictions were used directly. They were not rounded before the calculation.

7.3 CEFR Evaluation

CEFR was evaluated in both continuous and discrete forms.

The continuous ordinal score was

$$\hat{y}_{\text{soft}}^{(e)} = \sum_{k=1}^{5} \sigma(\ell_k). \tag{53}$$

Pearson correlation between this score and the integer CEFR target measured whether the learned proficiency axis preserved the correct ordering.

The uncalibrated hard prediction was

$$\hat{y}_{\text{hard}}^{(e)} = \sum_{k=1}^{5} I\left[\sigma(\ell_k) > 0.5\right]. \tag{54}$$

Both values lie between 0 and 5 and map to the ordered levels A1 through C2.

Exact CEFR accuracy was defined as

$$\text{Accuracy}_{\text{CEFR}} = \frac{1}{N_e} \sum_{i=1}^{N_e} I\left[\hat{y}_i^{(e)} = y_i^{(e)}\right]. \quad (55)$$

Only examples with an exact CEFR target were included in exact accuracy. Interval-labelled rows contributed to training but were not treated as exact validation labels.

7.4 Ordinal Distance Metrics

Exact accuracy does not distinguish a one-level error from a five-level error. Two additional ordinal metrics were therefore reported.

The mean CEFR distance was

$$D_{\text{mean}} = \frac{1}{N_e} \sum_{i=1}^{N_e} \left|\hat{y}_i^{(e)} - y_i^{(e)}\right|. \quad (56)$$

Lower values are better.

Within-one-level accuracy was

$$A_{\pm 1} = \frac{1}{N_e} \sum_{i=1}^{N_e} I\left[\left|\hat{y}_i^{(e)} - y_i^{(e)}\right| \leq 1\right]. \quad (57)$$

This metric is useful for ordinal proficiency assessment because confusing B1 with B2 is less severe than confusing A1 with C2.

7.5 Confusion Matrices

A 6×6 confusion matrix was generated for CEFR:

$$C_{i,j} = \#\left\{n : y_n = i \wedge \hat{y}_n = j\right\}. \quad (58)$$

Rows represent true levels and columns represent predicted levels. The matrix was inspected for:

- systematic overprediction or underprediction;
- collapse into one or two central levels;
- confusion between adjacent levels;
- failure to produce the highest or lowest levels; and
- separation between B2, C1, and C2.

Aggregate correlation can remain moderately high even when a model never predicts some classes. The confusion matrix was therefore necessary for interpreting CEFR behaviour.

7.6 Post-Training CEFR Calibration

The default hard prediction assumes that every ordinal probability should be compared with 0.5. Although this is natural for the binary threshold formulation, it does not guarantee the best discrete class boundaries on the validation distribution.

We therefore calibrated the mapping from the continuous CEFR score to the six labels. Calibration used only validation examples with exact CEFR labels.

For each adjacent class pair j and $j+1$, candidate cut-points were evaluated on the continuous scores of examples belonging to those two classes. The selected cut-point maximized binary separation accuracy:

$$c_j = \arg\max_c \frac{\sum_{i:y_i \in \{j,j+1\}} I\left[I[\hat{y}_{\text{soft},i}^{(e)} > c] = I[y_i = j+1]\right]}{\#\{i : y_i \in \{j, j+1\}\}}. \quad (59)$$

This produced five ordered cut-points:

$$c_0 < c_1 < c_2 < c_3 < c_4.$$

A calibrated prediction was then obtained by counting how many cut-points the soft score exceeded:

$$\hat{y}_{\text{cal}}^{(e)} = \sum_{j=0}^{4} I\left[\hat{y}_{\text{soft}}^{(e)} > c_j\right]. \quad (60)$$

The calibration parameters were stored in `cefr_calibration.json` together with the class mapping, adjacent-pair accuracies, number of calibration rows, and validation metrics.

7.7 Learned Thresholds and Calibration Cut-Points

The learned ordinal thresholds and post-training calibration cut-points serve different purposes.

The learned thresholds

$$\theta_1, \ldots, \theta_5$$

are trainable model parameters. They transform the learned latent proficiency score into five ordinal logits and are optimized through the CEFR training loss.

The calibration cut-points

$$c_0, \ldots, c_4$$

are fitted after training to the sum of the five ordinal probabilities. They do not alter the encoder, adapters, heads, or learned thresholds. They modify only the final conversion from the continuous score to a discrete CEFR label.

Consequently, calibration can change exact accuracy and mean class distance without changing the model's continuous CEFR correlation.

7.8 Validation Loss

Validation loss was calculated using the same masked objectives as training:

$$\mathcal{L}_{\text{val}} = \mathcal{L}_g + \mathcal{L}_c + \mathcal{L}_e. \quad (61)$$

No backward pass was performed. Dropout was disabled too.

The reported validation loss was the mean of the minibatch losses. It was used as a diagnostic quantity, while checkpoint selection relied on the composite metric in Equation 51.

7.9 Language-Wise Analysis

In addition to aggregate metrics, results were grouped by language. For each language, we calculated:

- number of examples;
- grammar Pearson correlation;
- content Pearson correlation;
- CEFR Pearson correlation; and
- CEFR exact accuracy.

This analysis was important because aggregate multilingual performance can hide poor behaviour in a specific language. It also helped identify the possible effect of having authentic CEFR augmentation for Dutch, Italian, and Portuguese but not for Greek and Polish.

Language-wise correlations were interpreted cautiously because each group contained considerably fewer examples than the complete evaluation set.

7.10 Shortcut Diagnostics

Several diagnostic baselines were evaluated on the controlled validation data before the final hybrid training experiments.

7.10.1 Length and Sentence-Count Baseline

A baseline using only character count, word count, and sentence count tested whether CEFR could be inferred from response length. Strong performance from this baseline would indicate that the synthetic construction had made longer responses systematically more advanced.

7.10.2 Grammar and Content Baseline

A second baseline used only the true grammar and content scores to predict CEFR. This measured how strongly the synthetic CEFR labels were coupled to the other two targets.

If this baseline were nearly perfect, the CEFR head could learn to reconstruct the synthetic label from grammar and content signals rather than learning broader proficiency characteristics.

7.10.3 TF–IDF Baselines

TF–IDF features were extracted from:

- question text only;
- email text only; and
- the concatenated question and email.

Simple linear predictors were used to assess how much label information could be recovered from lexical patterns. Separate TF–IDF diagnostics were also run for grammar and content.

The most strongly associated unigrams and bigrams were inspected for each CEFR level. Expressions that appeared almost exclusively in one class were treated as potential construction artifacts.

These baselines were not candidate final models. Their purpose was to detect unintended shortcuts in the generated dataset and motivate the addition of authentic learner writing.

7.11 Evaluation Principles

The evaluation followed four principles:

1. continuous predictions were retained for correlation metrics;
2. missing labels were excluded rather than converted into artificial targets;
3. CEFR was assessed using both exact and distance-sensitive metrics; and
4. aggregate results were supplemented with confusion matrices, language-level results, and shortcut diagnostics.

This combination was necessary because no single metric fully describes a three-task multilingual assessment system.

8 Results and Analysis

8.1 Overview of the Experiments

Four principal experiments were evaluated:

1. **Synthetic-5E**: controlled data, first-token pooling, and five epochs;
2. **Synthetic-8E**: controlled data, first-token pooling, and eight epochs;
3. **Hybrid-CLS**: controlled and authentic data, first-token pooling, and eight epochs; and
4. **Hybrid-Mean**: the same hybrid data with masked mean pooling.

Table 8 presents the selected checkpoint from each experiment before post-training CEFR calibration.

The synthetic and hybrid rows are not perfectly comparable because their validation compositions differ. The synthetic experiments were evaluated on the controlled validation subset, while the hybrid experiments additionally included authentic CEFR examples. Direct controlled comparisons are therefore most appropriate between Synthetic-5E and Synthetic-8E, and separately between Hybrid-CLS and Hybrid-Mean.

For the hybrid validation set, grammar and content metrics were calculated over 400 controlled examples with exact rubric labels. Exact CEFR metrics were calculated over 430 rows with exact CEFR labels. Interval-labelled examples contributed to CEFR training but were excluded from exact-label evaluation.

Table 8: Internal validation results for the four principal experiments. CEFR accuracy uses uncalibrated ordinal predictions.

Experiment	Validation loss	Grammar r	Content r	CEFR hard r	CEFR soft r	CEFR accuracy
Synthetic-5E	0.3613	0.7193	0.9573	0.8965	0.9096	0.4333
Synthetic-8E	0.3229	0.8693	0.9600	0.9161	0.9257	0.5500
Hybrid-CLS	0.2883	**0.9208**	0.9710	**0.8499**	**0.8702**	0.5977
Hybrid-Mean	**0.2868**	0.9063	**0.9767**	0.8354	0.8580	**0.6256**

8.2 Effect of Increasing the Training Duration

Synthetic-5E and Synthetic-8E used the same controlled data and the same first-token architecture. Increasing the training schedule from five to eight epochs improved grammar correlation from 0.7193 to 0.8693:

$$\Delta r_g = 0.8693 - 0.7193 = 0.1500. \tag{62}$$

Content correlation was already high after five epochs and changed only slightly:

$$\Delta r_c = 0.9600 - 0.9573 = 0.0027. \tag{63}$$

CEFR soft correlation increased from 0.9096 to 0.9257, while exact accuracy increased from 0.4333 to 0.5500. The validation loss also decreased from 0.3613 to 0.3229.

These results indicate that the five-epoch model had not fully converged, particularly for grammar. The content task was learned more quickly, probably because controlled content labels were closely tied to the presence or absence of complete response components.

8.3 Effect of Hybrid CEFR Data

The hybrid experiments incorporated authentic learner texts in addition to the controlled emails. Grammar and content correlation remained high because these labels continued to come from the controlled portion of the data.

The CEFR validation task became more heterogeneous. It now included both controlled emails and naturally occurring learner texts from several corpora. The authentic examples varied in domain, response length, error density, and annotation procedure. Some also had only interval supervision.

Compared with the synthetic validation experiment, the hybrid models showed lower CEFR correlation despite using more CEFR data. This does not necessarily indicate that the authentic data was harmful. The synthetic validation labels were partly associated with controlled grammar, content, and lexical patterns, making their ordering easier to predict. The hybrid validation set provided a more difficult and less regular proficiency-assessment problem.

The authentic data was therefore valuable primarily as a defence against learning a purely synthetic CEFR rule. It increased linguistic variety, although it also introduced a domain mismatch because much of the authentic material consisted of learner essays rather than prompt-conditioned professional emails.

Table 9: Change produced by replacing first-token pooling with masked mean pooling on the hybrid data.

Metric	CLS	Mean
Grammar r	**0.9208**	0.9063
Content r	0.9710	**0.9767**
CEFR hard r	**0.8499**	0.8354
CEFR soft r	**0.8702**	0.8580
CEFR accuracy	0.5977	**0.6256**
Validation loss	0.2883	**0.2868**

8.4 Pooling Ablation

Hybrid-CLS and Hybrid-Mean differed only in their sequence-pooling operation. Their comparison is shown in Table 9.

First-token pooling produced the stronger grammar correlation, with a difference of approximately 0.0145. Mean pooling produced the stronger content correlation, improving it by approximately 0.0057.

One possible interpretation is that the first contextual token acts as a learned global summary of the response, while mean pooling gives all non-padding positions a direct contribution. Content coverage depends on details distributed across the complete email, which may explain the improvement from averaging. Grammar quality may depend more on specific localized constructions, and uniform averaging can weaken the influence of a small number of informative error positions.

The CEFR results were mixed. First-token pooling produced higher hard and soft correlation, while mean pooling produced higher exact accuracy. This reflects a distinction between preserving the overall ordering and placing individual examples on the correct side of every class boundary.

The pooling experiment therefore did not identify one representation that dominated all three tasks. It instead suggested that the optimal sequence summary can depend on the target being predicted.

8.5 CEFR Calibration Results

Post-training CEFR calibration was performed separately for the two hybrid models. Table 10 compares their calibrated validation metrics.

For Hybrid-CLS, calibration improved exact accuracy from 0.5977 to 0.6674. For Hybrid-Mean, it improved accuracy from 0.6256 to 0.6767.

The fitted Hybrid-CLS cut-points were

$$[0.6938,\ 1.5756,\ 2.5953,\ 3.1156,\ 4.8008],$$

Table 10: CEFR calibration results on exact-label validation rows.

Model	Uncalibrated accuracy	Calibrated accuracy	Calibrated r	Mean distance	Within one level
Hybrid-CLS	0.5977	0.6674	**0.8569**	0.3698	**0.9674**
Hybrid-Mean	0.6256	**0.6767**	0.8522	**0.3674**	0.9581

Table 11: Adjacent CEFR separation for Hybrid-CLS.

Boundary	Pair accuracy
A1/A2	0.8403
A2/B1	0.8150
B1/B2	0.7841
B2/C1	0.8545
C1/C2	0.8644

Table 12: Adjacent CEFR separation for Hybrid-Mean.

Boundary	Pair accuracy
A1/A2	0.8750
A2/B1	0.8228
B1/B2	0.7709
B2/C1	0.8455
C1/C2	0.8814

while the Hybrid-Mean cut-points were

$$[0.6242,\ 1.5729,\ 2.2836,\ 3.4687,\ 4.5330].$$

These values differ from evenly spaced boundaries because the model's continuous score was not uniformly distributed across the six CEFR levels.

Calibration improved exact placement without retraining the model. However, these calibrated metrics were measured on the same validation partition used to fit the cut-points. They should therefore be interpreted as development-set results rather than as an independent estimate of calibration performance.

8.6 Adjacent-Level Calibration Analysis

The adjacent-class separation accuracies for Hybrid-CLS were:

The corresponding accuracies for Hybrid-Mean were:

For both models, B1/B2 was the most difficult adjacent boundary. This is consistent with the broad overlap between intermediate writing levels, where grammatical correctness, task completion, vocabulary, and structural complexity do not change at one exact point.

The strongest Hybrid-Mean boundaries were A1/A2 and C1/C2, while its B1/B2 separation remained below 0.80. This suggests that the model learned the extremes more clearly than the central proficiency transitions.

8.7 Synthetic-Data Shortcut Diagnostics

Before adding authentic learner texts, diagnostic baselines were used to measure whether CEFR labels could be recovered from superficial signals. The results are shown in Table 13.

The grammar-content baseline achieved a CEFR correlation of 0.920. This confirmed that the controlled CEFR labels were strongly associated with the other two rubric scores, even though their permitted ranges overlapped.

Table 13: CEFR shortcut diagnostics on the controlled data.

Input features	Accuracy	Pearson r
Length and sentence count	0.433	0.796
True grammar and content	0.533	0.920
Question text only	0.150	0.159
Email text only	0.417	0.710
Question and email text	0.450	0.647

The length-only correlation of 0.796 showed that response length and sentence count also carried substantial proficiency information. Although the C1 and C2 pairs had deliberately similar lengths, the complete A1–C2 ladder still became longer on average.

The email-only lexical baseline substantially outperformed the question-only baseline. This indicated that CEFR information primarily came from the candidate response, as intended, but also that recurring response expressions could identify synthetic levels.

TF–IDF regression produced correlations of approximately 0.612 for grammar and 0.861 for content. The especially high content result is understandable because content variants were generated by retaining or removing explicit task components. It also demonstrates that content coverage in the controlled data was lexically regular.

Inspection of class-associated unigrams and bigrams found recurring neutral fillers at lower levels and recurring cohesive or conditional phrases at higher levels. These diagnostics directly motivated the addition of authentic learner data. They also motivated varying prompt wording, greetings, closings, omitted components, and C1/C2 surface forms.

8.8 Training Behaviour of the Hybrid-CLS Model

The Hybrid-CLS model improved consistently across the eight-epoch schedule. Selected epoch-level results are shown in Table 14.

The largest gains occurred during the first four epochs. Improvements continued more gradually thereafter, particularly for grammar and exact CEFR accuracy. The composite selection score reached its highest value at epoch eight, so the final epoch was retained for this model.

There was no clear sign of severe overfitting within eight epochs. Training loss continued to decrease, while the primary validation metrics remained stable or improved. Nevertheless, the smaller changes after epoch six suggest that substantially longer training would probably provide diminishing returns.

Table 14: Selected validation results during Hybrid-CLS training.

Epoch	Grammar r	Content r	CEFR hard r	CEFR soft r	CEFR accuracy	Selection score
1	0.4404	0.8495	0.7125	0.7359	0.3372	0.5908
2	0.7038	0.9461	0.8055	0.8332	0.5000	0.7458
4	0.9084	0.9666	0.8423	0.8616	0.5605	0.8243
6	0.9158	0.9706	0.8419	0.8703	0.5581	0.8287
8	0.9208	0.9710	0.8499	0.8702	0.5977	0.8399

8.9 Selection of Checkpoints for Inference

The validation results show a task-dependent trade-off:

- Hybrid-CLS provided the strongest grammar correlation;
- Hybrid-Mean provided the strongest content correlation;
- Hybrid-CLS provided slightly stronger CEFR correlation; and
- Hybrid-Mean provided stronger exact CEFR accuracy after both uncalibrated and calibrated discretization.

This motivated retaining both hybrid checkpoints. A task-specific inference configuration can use the Hybrid-CLS grammar output and the Hybrid-Mean content output. For CEFR, the checkpoint can be selected according to the primary requirement. Hybrid-CLS is preferable if continuous correlation is prioritized, while Hybrid-Mean is preferable if exact level accuracy is prioritized.

For the packaged inference configuration, the chosen checkpoint and calibration file must be stated explicitly. Predictions from one checkpoint must not be combined with calibration cut-points fitted for the other checkpoint.

8.10 Summary of Findings

The principal findings were:

1. increasing controlled-data training from five to eight epochs substantially improved grammar and CEFR performance;
2. authentic CEFR data made the proficiency task more varied and reduced dependence on a purely controlled CEFR ladder;
3. first-token and mean pooling produced different task-specific advantages;
4. validation-based calibration improved exact CEFR placement without changing trained model parameters;
5. B1/B2 remained the most difficult adjacent CEFR boundary; and
6. simple diagnostic models exposed strong length, rubric, and lexical shortcuts in the controlled data.

All three prediction tasks exceeded the stated acceptance correlations on the internal validation partition, and CEFR exact accuracy exceeded the required threshold. However, the shortcut diagnostics and the difference between controlled and authentic writing indicate that the internal results should not be interpreted as evidence that the generalization problem has been completely solved.

9 Limitations and Risk Analysis

The system was developed under limited data, time, and computational resources. Although the final validation metrics are strong, several aspects of the data and methodology restrict the conclusions that can be drawn from them.

9.1 Synthetic Structure and Repeated Scenarios

The controlled portion contains 2,000 examples, but these examples are derived from only 40 underlying scenarios. Multiple content, grammar, and CEFR variants therefore share the same factual structure.

The scenario-level split prevents direct leakage of one scenario between training and validation. Nevertheless, the model sees repeated patterns such as requesting an action, providing supporting information, and asking for confirmation. This is appropriate for professional email writing, but it limits the diversity of discourse structures represented in the data.

The five task components also impose a relatively regular organization:

$$\text{purpose} \rightarrow \text{context} \rightarrow \text{action} \rightarrow \text{support} \rightarrow \text{confirmation}.$$

The order was varied in some generated responses, and different components were omitted for intermediate content scores. Even so, the controlled examples remain more regular than naturally written emails.

9.2 Content Labels Are Programmatically Defined

Content labels were assigned from the number of task components retained in the response. This produces clear and reproducible labels, but it is only an approximation of human content assessment.

In real writing, details differ in importance. Omitting the central requested action may be more serious

than omitting a reference to an attachment. A partially expressed detail may also deserve partial credit. The atomic-detail schema and importance weights were stored to support this analysis, but the final content labels were generated from broad component counts.

The controlled formulation therefore measures structured coverage more directly than it measures every aspect of human relevance, adequacy, or communicative success.

9.3 Programmed Grammar Errors

The grammatical perturbations were typed and language-aware, but they were still generated by deterministic rules. They do not represent the full distribution of errors made by language learners.

Some error categories occurred much more frequently than others. Function-word omissions, article substitutions, missing diacritics, and minor spelling changes were easier to apply across many sentences. More context-dependent phenomena, such as lexical selection, idiomaticity, discourse agreement, tense consistency, and long-range syntactic errors, were less well represented.

A generated operation can also be linguistically awkward without matching a naturally occurring learner error. Although unchanged rows, inverse operations, and protected-token damage were checked automatically, the complete 1,800-row grammar dataset was not rated independently by native speakers.

The grammar labels should therefore be understood as controlled rubric approximations rather than gold-standard human judgements.

9.4 Unequal Error Coverage Across Languages

The availability of a particular grammatical rule depended on whether its required pattern occurred in a clean response. For example, a Polish case-selection rule can only be applied when a matching noun phrase is present, and a Dutch word-order rule requires a compatible clause.

As a result, error-type frequencies were not identical across languages. Polish received more case and preposition errors, Dutch received article and verb-position errors, and languages with diacritics received missing-mark errors. This reflects genuine linguistic differences, but it also means that the grammar task was not perfectly matched across languages.

Language-wise performance should therefore be inspected alongside the aggregate grammar correlation.

9.5 Synthetic CEFR Anchors

The A1–B2 anchors were chosen conservatively from overlapping grammar and content ranges. This was preferable to labelling every generated example, but the selected labels were not produced by certified CEFR raters.

The C1 and C2 examples were separately formulated to express greater lexical precision, cohesion, and syntactic control. Their lengths were kept similar to avoid a direct C1/C2 length shortcut. However, both groups used clean grammar and complete content. This creates a relationship between advanced CEFR, grammar score 5, and content score 4.

The synthetic diagnostic baselines confirmed that grammar, content, length, and recurring expressions remained predictive of CEFR. The addition of authentic data reduced reliance on this controlled ladder but could not completely remove patterns learned from the synthetic portion.

9.6 Limited Representation of Short Advanced Writing

A concise email can demonstrate C1 or C2 proficiency through precision, register, and efficient expression. The controlled dataset includes similar-length C1 and C2 pairs, but lower levels are shorter on average than upper levels across the complete A1–C2 range.

Consequently, the model may partially associate length with proficiency. The length-only baseline confirmed that this signal was substantial. Future data should include more short advanced responses and more verbose lower-level responses while controlling the facts communicated.

9.7 Authentic-Data Domain Mismatch

The authentic CEFR corpora contain naturally occurring learner language, but they do not match the target task exactly. Many examples are essays or general written responses rather than professional emails written in response to structured prompts.

These rows had an empty prompt and contributed only to CEFR training. They increased linguistic realism but introduced a domain difference between:

- authentic, longer, prompt-independent learner texts; and
- controlled, shorter, prompt-conditioned emails.

The model may therefore learn source-related distinctions in addition to proficiency. A stronger dataset would contain authentic prompt-response email pairs annotated jointly for grammar, content, and CEFR.

9.8 Language Imbalance in the Authentic Data

The authentic augmentation included Dutch, Italian, and Portuguese, but no corresponding Greek or Polish learner corpus was incorporated. The four selected subsets contained:

- 645 Dutch texts;
- 600 Italian texts;
- 755 Portuguese texts across two sources; and
- no authentic Greek or Polish texts.

Thus, the CEFR head received naturally occurring learner supervision for only three of the five target languages. XLM-RoBERTa can transfer features across languages, but cross-lingual transfer cannot be assumed to remove this imbalance completely.

9.9 Portuguese Variety

The Portuguese data combined controlled professional emails with learner corpora that may contain different Portuguese varieties. Some controlled expressions are more characteristic of Brazilian Portuguese, while public learner corpora can be oriented toward European Portuguese.

Both are valid forms of Portuguese, but mixing them without an explicit variety label can introduce variation in vocabulary, pronouns, prepositions, and register. A future version should either standardize the target variety or record it as metadata.

9.10 LLM-Assisted Localization and Construction

Language-model assistance was used while drafting multilingual scenario content, language-specific variants, parts of the data generation code, and supporting documentation. The localized records were explicitly stored and checked for preservation of scenario facts, identifiers, task requirements, and formatting.

However, this process is not equivalent to independent annotation by native speakers. The label `manually_curated_multilingual` refers to manual inspection and organization of the records, not professional human translation.

This distinction is stated explicitly so that the provenance of the controlled data is not overstated.

9.11 Validation Reuse

The internal validation partition served several purposes:

- checkpoint selection;
- comparison of model variants;
- CEFR cut-point calibration; and
- reporting of final development metrics.

Repeated use of one validation partition can lead to indirect overfitting, even when no gradients are computed from it. In particular, calibrated CEFR accuracy was measured on the same examples used to select the cut-points.

The calibrated results should therefore be interpreted as development performance. A stronger evaluation would fit calibration on a separate calibration partition and report final metrics on a third untouched partition.

9.12 Simple Pooling Operations

Only two sequence-level representations were tested:

- the first contextual token; and
- the attention-mask-aware mean of token representations.

Mean pooling treats all non-padding positions equally, even though some tokens are more informative than others. First-token pooling relies on one position to summarize the complete sequence.

An attention-pooling layer, hierarchical email representation, or separate prompt and response encoders might capture task-relevant information more effectively. These alternatives were not explored because the goal was to keep the model small, reproducible, and trainable on one T4 GPU.

9.13 Restricted Fine-Tuning Search

LoRA was applied only to the query and value projections with rank eight. The experiment did not include a broad search over:

- LoRA rank or scaling;
- key, output, or feed-forward adapters;
- full-model fine-tuning;
- task-loss weights;
- alternative learning rates;
- focal or robust regression losses;
- longer training schedules; or
- larger multilingual encoders.

The selected configuration is a practical parameter-efficient baseline, not evidence that the architecture is globally optimal.

9.14 Equal Multitask Loss Weights

Grammar, content, and CEFR losses were summed with equal numerical weight. Their effective influence can still differ because they have different numbers of labelled examples, different noise levels, and different gradient magnitudes.

The strong content performance suggests that content may be easier to learn from the controlled construction. CEFR is more heterogeneous and may require a different loss weight or sampling strategy. Dynamic task weighting and gradient-conflict methods were outside the scope of the experiments.

9.15 Input Truncation

The maximum sequence length was 384 subword tokens. Most controlled emails and most authentic responses fit within this limit, but a small fraction of the longest learner texts were truncated.

Truncation can remove conclusions, later grammatical errors, or evidence of discourse organization. Increasing the maximum length or using chunk-based aggregation would preserve more information, at the cost of additional computation.

9.16 Calibration Under Distribution Shift

The CEFR calibration cut-points were fitted to the internal validation score distribution. If deployment data has different languages, domains, response lengths, or proficiency proportions, these cut-points may no longer be optimal.

For this reason, the inference pipeline should retain both:

- the continuous CEFR score; and
- the final calibrated CEFR label.

The continuous score makes it possible to inspect uncertainty and recalibrate the model when representative labelled data becomes available.

9.17 Overall Interpretation

The model demonstrates that controlled email supervision, authentic CEFR data, ordinal learning, and parameter-efficient multilingual adaptation can be combined in one reproducible pipeline. The internal results confirm that the implementation learns all three objectives.

They do not establish that the system is a replacement for expert language assessment. Its scores are affected by programmed errors, synthetic structure, domain differences, language imbalance, and validation reuse. The appropriate interpretation is that the system is a technically grounded prototype whose remaining weaknesses are identifiable and testable.

10 Inference Configuration

The two hybrid models showed complementary validation performance. Hybrid-CLS produced the highest grammar correlation, while Hybrid-Mean produced the highest content correlation and exact CEFR accuracy. The final inference configuration therefore uses:

- Hybrid-CLS for grammar prediction;
- Hybrid-Mean for content prediction; and
- Hybrid-Mean with its corresponding calibration cut-points for CEFR prediction.

The inference program accepts a CSV or Excel file containing `questionStatement` and `text`. It tokenizes each prompt-response pair using the saved XLM-RoBERTa tokenizer and applies the same truncation and dynamic-padding procedure used during training.

The output contains a continuous grammar score in $[0, 5]$, a continuous content score in $[0, 4]$, a continuous ordinal CEFR score, and a calibrated CEFR label from A1 to C2. The original row identifiers and row order are preserved.

Inference is exposed through a command-line interface:

```
python src/inference.py \
    --input test_dataset.xlsx \
    --output predictions.csv
```

Detailed installation instructions, checkpoint locations, input and output schemas, dependency versions, and model-reloading commands are provided in the repository README.

11 Conclusion

This work developed a multilingual multitask system for predicting grammar, content coverage, and CEFR proficiency from an email prompt and response. The training data combined controlled multilingual emails with authentic CEFR-labelled learner texts.

The model used XLM-RoBERTa with LoRA, bounded regression heads for grammar and content, and an ordinal head for CEFR. Four experiments examined training duration, authentic-data augmentation, and pooling. First-token pooling performed best for grammar, while mean pooling performed best for content and exact CEFR accuracy.

The main limitations are the structured nature of the synthetic data, the domain difference between learner corpora and professional emails, and the absence of authentic Greek and Polish data. Future work should prioritize authentically written emails jointly annotated for all three tasks.

AI-Assistance Disclosure

Large language model assistance was used for brainstorming, code scaffolding and debugging, multilingual drafting, and documentation editing. The author was responsible for the initial model and dataset designing ideas, final design choices, implementation, execution, validation, and interpretation of results. The controlled multilingual data is therefore described as LLM-assisted and manually inspected, not as professionally translated or independently human-annotated.